Lectures 3+4: Conditional probability (Ch. 2)

Motivating example (The Lottery Ticket example).

- Lottery game: 6 numbers are drawn, without replacement, from a bin containing $\{1, 2, \ldots, 30\}$.
- Your ticket = 6 numbers.
- You win: your 6 numbers are the 6 numbers drawn (on TV), without regard to the order.
- You turn on the TV and don't see the whole show, but only that, say, 15 was drawn.
- Say your ticket is $A = \{1, 14, 15, 20, 23, 27\}$.
- Since you have seen 15 being drawn, the probability that you win must now be higher than before the draw.

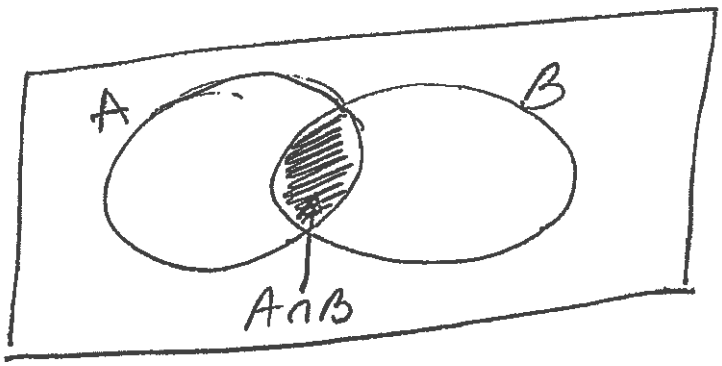
How to quantify this ???

Definition (Conditional probability)

If the event B has occured, the conditional probability of the event A given that B has occured is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad \underline{\text{if}}$$

$P(B) > 0$.



$P(A|B)$ = "The proportion of B that is also part of A."

Solution to the "Lottery ticket problem".

We want $P(A|B)$, where

B = "Any 5 numbers + the number 15."

- Sample space = All 6 unordered draws of six numbers from 1 to 30.

$=: S.$

$|S| = \binom{30}{6}$, all possible combinations of 30 numbers taken 6 at a time.

- S has equally likely outcomes.

- The event B: all combinations of 6 numbers, one of which must be 15, that can be selected out of 30 numbers.

$$P(B) = \frac{\binom{29}{5}}{\binom{30}{6}}$$

$\binom{29}{5}$ ← all combinations of 5 distinct numbers out of 29.

- $$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\cdot\ P(A \cap B) = P(A) = \frac{1}{\binom{30}{6}} \doteq 1.68 \times 10^{-6}$$

Therefore:

$$P(A|B) = \frac{1}{\binom{30}{6}} \cdot \frac{\binom{30}{6}}{\binom{29}{5}} = 8.4 \times 10^{-6},$$

and so this is 5 times larger than $P(A)$ before you learned that B had occured.

Solution 2: Working on the "reduced sample space".

- Since you've seen 15 being drawn, the only uncertainty comes from selecting the remaining 5 numbers out of the "reduced bin", of 29 numbers.
- Since "win = one particular configuration is being drawn", then

$$\mathbb{P}(\text{"Win"} \mid \text{"you've seen 15"}) = \frac{1 \leftarrow \text{one possible configuration}}{\binom{29}{5} \leftarrow \text{total number of configurations}}$$

The Multiplication rule for conditional probabilities

Theorem 1. • If $P(B)>0$: $P(A\cap B)=P(A|B)\cdot P(B)$.

• If $P(A)>0$: $P(A\cap B)=P(B|A)\cdot P(A)$.

Example Two balls are to be selected at random, without replacement, from a box with 10 red balls and 7 blue balls. What is:

$$P(\text{First ball red, second ball blue}) =$$

$$= P(\{\text{First ball red}\}\cap\{\text{Second ball blue}\})$$

$$= P(\{\text{First ball red}\})\cdot P(\{\text{Second blue}\}|\{\text{First red}\})$$

$$= \frac{10}{10+7}\cdot\frac{7}{(10-1)+7} = \frac{10}{17}\cdot\frac{7}{16}$$

Theorem 2 Let $A_1, \ldots, A_n$ be events such that $P(A_1 \cap A_2 \cap \ldots \cap A_{n-1}) > 0$. Then:

$$P(A_1 \cap A_2 \cap \ldots \cap A_n) = P(A_1) \cdot P(A_2 | A_1) \cdot P(A_3 | A_1 \cap A_2) \cdot \ldots \cdot P(A_n | A_1 \cap A_2 \cap \ldots \cap A_{n-1}).$$

NOTE Assume that an event (or intersection of events) has occured. Then, all the results stated before in terms of probabilities can be re-phased in terms of conditional probabilities conditional on the event(s) that have occured.

For instance, if $P(B) > 0$ then

$$P(A^c | B) = 1 - P(A | B).$$

Reading assignment: Th. 2.1.3 page 59 book.

Partitions of the sample space and conditional probabilities

Definition Let S = sample space. Let $B_1, \ldots, B_K$ be K events in S such that $B_1, \ldots, B_K$ are disjoint and $\bigcup_{i=1}^{K} B_i = S$. if it is said that these events form a partition of S.

Theorem 3 (Law of total probability). Assume $B_1, \ldots, B_K$ form a partition of the sample space. Then, for every event A in S:

$$P(A) = \sum_{j=1}^{K} P(B_j) \cdot P(A|B_j),$$

provided that $P(B_j) > 0$, for $j = 1, \ldots, K$.

Example

Box 1 : 60 long bolts + 40 short bolts.

Box 2 : 10 long bolts + 20 short bolts.

A box is selected at random and then a bolt.

$P(\text{"The selected bolt is long"}) = ?$ (A)

Solution : . Let B_1 be the event that Box 1 is selected.

. Let B_2 " " " " Box 2 " " .

$$P(A) = P(A|B_1) \cdot P(B_1) + P(A|B_2) \cdot P(B_2)$$

$$= \frac{60}{100} \cdot \frac{1}{2} + \frac{10}{30} \cdot \frac{1}{2} = \frac{7}{15}.$$

Theorem 4 (Conditional version of Theorem 3)

$$P(A|C) = \sum_{j=1}^{K} P(B_j|C) \cdot P(A|B_j \cap C).$$

Independent Events

Independence of two events

Definition Two events A and B are independent if

$$P(A \cap B) = P(A) \cdot P(B).$$

Important remark Suppose A and B are s.t. $P(A) > 0$ and $P(B) > 0$.

A and B are independent iff

$P(A|B) = P(A)$ and $P(B|A) = P(B)$.

Example 1. • Machines 1 and 2 are operated independently of each other.

• A = Machine 1 will stop; $P(A) = \frac{1}{3}$

B = Machine 2 will stop; $P(B) = \frac{1}{4}$.

• P("At least one machine stops")

$= P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\overset{\uparrow}{=} P(A) + P(B) - P(A) \cdot P(B)$

Independence

$= \frac{1}{3} + \frac{1}{4} - \frac{1}{12} = \frac{1}{2}$.

Example 2 (Events may be independent, even though they may be physically related).

• A die is rolled.

A = an even number is obtained.

B = One of the numbers 1, 2, 3 or 4 is obtained.

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

$$P(B) = \frac{4}{6} = \frac{2}{3}$$

$$P(A \cap B) = P(\{2, 4\}) = \frac{2}{6} = \frac{1}{3}$$

Since $P(A \cap B) = \frac{1}{3} = P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$,

A and B are independent.

<u>Result</u> If A and B are independent, then the following events are also independent:

(a) A and B^c,

(b) A^c and B,

(c) A^c and B^c.

Independence of several events

Definition **Mutually independent events.**

The k events are independent (or mutually independent) if, for every subset $A_{i_1}, \ldots, A_{i_j}$ of j of these events ($j = 2, 3, 4, \ldots$), we have

$$\circledast \quad P(A_{i_1} \cap \ldots \cap A_{i_j}) = P(A_{i_1}) \cdot \ldots \cdot P(A_{i_j}) .$$

Remark If the equality $\circledast$ above is only required to hold for $j = 2$, that is:

$$P(A_{i_1} \cap A_{i_2}) = P(A_{i_1}) \cdot P(A_{i_2}),$$ for every pair of the k events, then the events are called pairwise independent.

NOTE PAIRWISE INDEPENDENCE $\not\Longrightarrow$ INDEPENDENCE.

- Independence of 3 events

A, B and are called (mutually) independent if:

$$\left.\begin{aligned} P(A\cap B) &= P(A)\cdot P(B)\\ P(A\cap C) &= P(A)\cdot P(C)\\ P(B\cap C) &= P(B)\cdot P(C) \end{aligned}\right\} \quad (1)$$

and

$$P(A\cap B\cap C) = P(A)\cdot P(B)\cdot P(C) \quad (2)$$

- Example of 3 events that are pairwise independent (① holds), but not mutually independent (② does not hold).

- Let $S = \{HH, HT, TH, TT\}$ be associated with tossing a coin twice.

$A = \{H$ on first toss$\} = \{HH, HT\}$

$B = \{H$ on second toss$\} = \{HH, TH\}$

$C = \{$Both tosses the same$\} = \{HH, TT\}$.

- $A\cap B = A\cap C = B\cap C = \{HH\} \Rightarrow$

$$\frac{1}{4} = P(A\cap B) = P(A)\cdot P(B) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$$

$$\frac{1}{4} = P(A\cap C) = P(A)\cdot P(C) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$$

$$\frac{1}{4} = P(B\cap C) = P(B)\cdot P(C) = \frac{1}{4}$$

· However, $A \cap B \cap C = \{HH\} \Rightarrow$

$P(A \cap B \cap C) = \frac{1}{4}$, but

$P(A) \cdot P(B) \cdot P(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \neq \frac{1}{4}$.

Example • A machine produces, independently:
- defective items, with probability p,
- nondefective items, with probability $1-p$.

• Six items produced by this machine are selected.

P("At least one of the 6 items is defective") = ?

P("At least one of the 6 items is defective") = $[\text{None}]^c$.

NB In general, $[\text{At least one}] = [\text{None}]^c$.

P("At least one is defective") $= 1 - P$("None is defective")

$= 1 - P[$"Item 1 is nondefective" A_1 and ... "Item 6 is nondef." A_6"]

$= 1 - P[A_1 \cap A_2 \cap \ldots \cap A_6]$

$\overset{4}{=} 1 - P(A_1) \cdot P(A_2) \cdot \ldots \cdot P(A_6) = 1 - (1-p)^6$.

Independence

IMPORTANT Do not confuse

"Mutually exclusive" with "Mutually independent"

$\Updownarrow$ $\Updownarrow$

$A \cap B = \emptyset$ $P(A \cap B) = P(A) \cdot P(B)$

Reading assignment: Th. 2.2.3 page 72 book.

Conditional independence (ci)

Definition A_1 and A_2 are conditionally independent given B if

$$P(A_1 \cap A_2 | B) = P(A_1|B) \cdot P(A_2|B).$$

Result Let A_1, A_2 and B such that $P(A_1 \cap B) > 0$.

Then:

A_1 and A_2 are ci given B iff $P(A_2 | A_1 \cap B) = P(A_2|B)$

Example

- You want to compile 11 similar programs.
- A_i = the i-th program compiles successfully; $i = 1, 2, \ldots, 11$.
- B = the programming task is easy
- $P(\text{"Program compiles"} \mid \text{"Easy"}) = 0.8$
  $P(\text{"Compiles"} \mid \text{"Hard"}) = 0.4$

Assume: • $A_1, \ldots, A_{11}$ conditionally independent given B

- $A_1, \ldots, A_{11}$ conditionally independent given B^c.

(a) $P(\text{Exactly 8 out 11 compile} \mid B) = ?$

(b) $P(\text{" " " " " " } \mid B^c) = ?$

(a) $\textcircled{**} = P(\text{"Exactly 8 out of 11 compile"} \mid \text{Easy})$

$$= \sum_{\{\text{All possible ways of chosing 8 programs out of 11}\}} \underbrace{P(\text{"A particular configuration of 8 programs out of 11 compiles"} \mid \text{Easy})}_{\textcircled{*}}$$

- Let $i_1, \ldots, i_8$ be the numbers of the programs that will compile.

Then

$$\textcircled{*} = P(A_{i_1} \cap A_{i_2} \cap \ldots \cap A_{i_8} \cap A_{i_9}^c \cap A_{i_{10}}^c \cap A_{i_{11}}^c \mid B)$$

$$= \prod_{j=1}^{8} P(A_{i_j} \mid B) \times \prod_{j=9}^{11} P(A_{i_j}^c \mid B)$$

↑ Conditional independence

$$= (0.8)^8 \cdot (0.2)^3 = 0.001342.$$

- # of ways of chosing 8 out 11 is

$$\binom{11}{8} = 165.$$

Therefore :

$$P(\text{"Exactly 8 out of 11 compile"} \mid \text{Easy}) = 165 \times 0.001342 = 0.2215$$

(b) Reasoning similarly :

$$P(\text{"Exactly 8 out of 11 compile"} \mid \text{Hard}) = 165 \times (0.6)^8 \times (0.4)^3 = 0.02335.$$

Note (a) $\neq$ (b) !

Bayes' Theorem

General framework

- We are interested in the occurrence of k disjoint events $B_1, \ldots, B_k$.
- Let A be an event that may be informative relative to $B_1, \ldots, B_k$.
- Assume we have access to $P(A|B_i)$, for each i.
- Bayes' Theorem can be used to combine this information to compute $P(B_i|A)$, for each i.

Example

B_1 = Jones has the disease
B_2 = Jones doesn't have the disease } disjoint

A = positive test result

Compute $P(B_1 | A)$. //

$$P(B_1 | A) = \frac{P(B_1 \cap A)}{P(A)}$$

Suppose that we have the following information:

- $P(B_1) = 0.6$ ← This can be regarded as "the personal" probability of the doctor, based on clinical observation, but not on tests.
- $P(A | B_2) = 0.3$ ← The test yields "false positives" 30% of the time
- $P(A | B_1) = 1$.

By the "Law of total probability" (Th. 3 page 43 notes)

$$P(A) = P(A|B_1)\cdot P(B_1) + P(A|B_2)\cdot P(B_2);$$

Note that $B_2 = B_1^c$. Therefore, $P(B_2) = 1 - 0.6 = 0.4$

Then $P(A) = 1 \times 0.6 + 0.3 \times 0.4$

$$= 0.6 + 0.12 = 0.72$$

Also, by the multiplication rule:

$$P(B_1 \cap A) = P(A|B_1)\cdot P(B_1) = 1 \times 0.6 = 0.6$$

Therefore:

$$P(B_1|A) = \frac{P(B_1)\cdot P(A|B_1)}{P(B_1)\cdot P(A|B_1) + P(B_2)\cdot P(A|B_2)}$$

$$= \frac{0.6}{0.72} = 0.833.$$

Theorem (Bayes' Theorem).

Let $B_1, \ldots, B_K$ form a partition of the space S such that $P(B_j) > 0$, for $j = 1, 2, \ldots, K$. Let A be an event such that $P(A) > 0$. Then, for every $i = 1, \ldots, K$,

$$P(B_i \mid A) = \frac{P(B_i) \cdot P(A \mid B_i)}{\sum_{j=1}^{K} P(B_j) \cdot P(A \mid B_j)}$$

Example

- Machines M_1, M_2 and M_3 produce similar items.
- An item is selected at random from a batch.
- B_i = the event that the selected item was produced by machine M_i; $i = 1, 2, 3$; A = the selected item is defective.
- Given: $P(B_1) = 0.2$ (M_1 produces 20% of items).

  $P(B_2) = 0.3$ (M_2 produces 30%)

  $P(B_3) = 0.5$ (M_3 produces 50%).

  $P(A|B_1) = 0.01$ (1% of items of M_1 are defective)

  $P(A|B_2) = 0.02$

  $P(A|B_3) = 0.03$.
- Want $P(B_2|A)$ = Prob. that the selected item was produced by M_2.

· Bayes' Theorem ⟹

$$P(B_2|A) = \frac{P(B_2)\cdot P(A|B_2)}{\sum_{j=1}^{3} P(B_j)\cdot P(A|B_j)}$$

$$= \frac{0.3 \times 0.02}{0.2\times 0.01 + 0.3\times 0.02 + 0.5\times 0.03}$$

$$= 0.26 .$$

Prior and posterior probabilities

In the example above, any of the probabilities $P(B_1)$, $P(B_2)$ or $P(B_3)$ can be called prior probabilities: This is the probability of (any of) these events before an item was selected and before we had any extra information about the selected item (defective or not).

- For instance, $P(B_2)$ is the prior probability that the selected item will have been produced by M_2

- Then, $P(B_2|A)$ is called the posterior probability that the selected item was produced by M_2, since this is calculated after we know that it is defective.

In general: We use Bayes' theorem to update our initial beliefs in light of new evidence (data).

- Notice that in the previous example,

$P(B_2|A) < P(B_2)$.

- Is it always the case that $P(B_j|A) < P(B_j)$?

NO! See next page.

Result Let $B_1, \ldots, B_K$ form a partition of S. Let A be an event with $P(A) > 0$.

If $P(B_1|A) < P(B_1)$, then $P(B_i|A) > P(B_i)$, for at least one value of $i = 2, 3, \ldots, K$.

Proof $\{B_1, \ldots, B_K\}$ is a partition of $S \Rightarrow$

$$\sum_{i=1}^{K} P(B_i) = 1 \text{ and}$$

$$\sum_{i=1}^{K} P(B_i|A) = 1.$$

Assume that the conclusion is not true, that is, assume that $P(B_1|A) < P(B_1)$

and

$$P(B_i|A) < P(B_i), \quad i = 2, 3, \ldots, K.$$

Summing up LHS and RHS: $1 = \sum_{i=1}^{K} P(B_i|A) < \sum_{i=1}^{K} P(B_i) = 1 \Rightarrow$

$1 < 1$, contradiction!

Therefore, there exists at least one $i \geq 2$ such that $P(B_i|A) > P(B_i)$.